

PBL PROJECT REPORT

AI-BASED HANGOUT RECOMMENDATION SYSTEM USING CHAT SENTIMENT ANALYSIS

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Submitted by:

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ABSTRACT

This project presents an AI-driven Hangout Recommendation System that analyzes the mood and sentiment of conversations between two or more individuals. Using sentiment analysis and Natural Language Processing (NLP), the system identifies whether a group chat conveys a positive, negative, or neutral mood. Based on the detected sentiment, it recommends suitable hangout places such as cafés, malls, nature spots, or quiet parks. The model is trained using the Sentiment140 Kaggle dataset, which contains 1.6 million labeled tweets. A lightweight TF-IDF + Logistic Regression classifier is used to predict sentiment efficiently. The output is combined with a rule-based place suggestion engine to provide relevant, mood-appropriate recommendations.

INTRODUCTION

Artificial Intelligence (AI) is increasingly used to enhance decision-making and personalization. Group outings and hangout planning depend heavily on the collective mood

or vibe of the participants. This project introduces a simple yet effective system that automatically reads a conversation, identifies its emotional tone, and recommends places that best match the overall mood.

OBJECTIVES

- To understand and implement sentiment analysis using Machine Learning.
- To train a sentiment classifier using the Sentiment140 dataset.
- To analyze group conversations and detect dominant mood.
- To recommend places that match the sentiment.
- To create a simple real-world AI system.

DATASET DETAILS

Dataset Used: Sentiment140

Dataset Size: 1.6 million tweets

Labels: 0 = Negative, 4 = Positive

METHODOLOGY

1. Data Preprocessing
2. TF-IDF Feature Extraction
3. Logistic Regression Training
4. Chat Mood Analysis
5. Rule-Based Recommendation Engine

IMPLEMENTATION

Technologies Used:

Python, Scikit-learn, Pandas, TF-IDF, Logistic Regression, Google Colab

RESULTS AND DISCUSSION

The system detects mood from group chat messages and recommends relevant hangout places based on positive, negative, or neutral sentiment.

CONCLUSION

This PBL demonstrates a real-world AI application using NLP and sentiment analysis. The system successfully recommends places based on group chat mood.

FUTURE WORK

- WhatsApp/Instagram chat integration
- Full emotion analysis
- Web UI Deployment
- GPS-based recommendations

REFERENCES

1. Sentiment140 Dataset — Kaggle
2. Scikit-Learn Documentation
3. NLP with Python (Book)
4. TF-IDF Documentation
5. Logistic Regression — Stanford CS229